HRITIKA SHARMA

AI Evaluation | Machine Learning Research | Data Analytics

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PROFESSIONAL SUMMARY

Computer Science (Honours) graduate with experience in production AI evaluation, human-aligned speech-recognition research and data analytics. First author of an ASR evaluation preprint; uses Python, SQL and Pandas to validate data, investigate model-evaluation discrepancies and deliver reproducible analyses of datasets exceeding 7.8 million records.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, PostgreSQL, MySQL, Excel

AI & Research: NLP/ASR evaluation, rubric-based assessment, annotation quality, error analysis, data validation

Analytics & Tools: Power BI, Power Query, DAX, Matplotlib, Git/GitHub, Jupyter, VS Code

PROFESSIONAL EXPERIENCE

Undergraduate Research Assistant -- Algoma University | Ontario | May 2025–Sep 2026

- Validated human-annotated ASR research data using Python and Pandas, identifying missing values, inconsistent records and anomalous evaluation results for investigation.
- Compared automated ASR metrics with human judgments, identifying cases where numerical scores missed perceived transcription quality; first-authored the resulting research preprint (2026).

AI Evaluation Contributor -- Project Hedgehog, Handshake AI | Remote | 2026

- Completed accepted production evaluations of AI-generated outputs against task-specific rubrics, documenting quality defects and evidence-based judgments to support consistent review.

IT Intern -- Innovation Ghar Nepal | Kathmandu, Nepal | Jun 2021–Feb 2022

- Troubleshot computers, software and robotics training equipment to support workshop readiness; prepared instructional materials for introductory programming sessions.

RESEARCH PUBLICATION

Rethinking Human-Aligned Evaluation: An Analysis of Semantic Metrics Beyond WER | First-author arXiv preprint | 2026 | View Paper

- Investigated disagreements between Word Error Rate, semantic evaluation metrics, and human judgments to examine limitations in automated ASR transcription evaluation.

SELECTED PROJECTS

Toronto 311 Service Request Analysis | Python, SQL, PostgreSQL, Power BI

- Repaired 36,189 malformed records within 500,269 service requests while preserving all rows, enabling complete downstream SQL analysis and a two-page operational Power BI dashboard.
- Used CTEs, window functions and analytical views to examine service demand by time period, division, request type, ward and status.

Toronto Bike Share 2025 Analysis | Python, Pandas, Matplotlib

- Analyzed 7.8M+ trips over 12 months to identify seasonal and member-versus-casual usage patterns; investigated daylight-saving-related timestamp discrepancies before interpreting trip durations.

English - Nepali AI Response Evaluation Pilot | Python, Pandas, rubric design

- Designed a structured rubric and reviewed 10 responses across multilingual tasks, identifying instruction-following and completeness errors; documented the small-sample limitations.

EDUCATION

Bachelor of Science (Honours), Computer Science | Algoma University | Brampton, ON | Jun 2026